

Symptomatic efficacy of avocado-soybean unsaponifiables (ASU) in osteoarthritis (OA) patients: a meta-analysis of randomized controlled trials.

OBJECTIVE: To evaluate the efficacy of preparations with avocado-soybean unsaponifiables (ASUs) in osteoarthritis (OA) patients using meta-analysis on randomized controlled trials (RCTs). **METHOD:** RCTs from systematic searches were included if they explicitly stated that hip and/or knee OA patients were randomized to either ASU or placebo. The co-primary outcome was reduction in pain and Lequesne index, leading to effect size (ES), calculated as the standardized mean difference. As secondary analysis, the number of responders to therapy was analyzed as odds ratios (ORs). Restricted maximum likelihood methods were applied for the meta-analyses, using mixed effects models. **RESULTS:** Four trials--all supported by the manufacturer--were included, with 664 OA patients with either hip (41.4%) or knee (58.6%) OA allocated to either 300 mg ASU (336) or placebo (328). Average trial duration was 6 months (range: 3-12 months). Though based on heterogeneous results, the combined pain reduction favored ASU ($I(2) = 83.5\%$, $ES = 0.39$ [95% confidence intervals: 0.01-0.76], $P=0.04$). Applying the Lequesne index also favored ASU ($I(2) = 61.0\%$, $ES = 0.45$ [0.21-0.70], $P = 0.0003$). Secondly, the number of responders following ASU compared to placebo ($OR = 2.19$, $P = 0.007$) corresponded to a number needed to treat of six (4-21) patients. **CONCLUSIONS:** Based on the available evidence, patients may be recommended to give ASU a chance for e.g., 3 months. Meta-analysis data support better chances of success in patients with knee OA than in those with hip OA.